

Boundary Value Problems in Electrostatics

PHYS505 – Electromagnetic Theory

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Method of images

- Special trick that happens to work in some problems with suitable geometry/symmetry
- *Image charges* placed external to the volume of interest result in identical boundary conditions to the problem in hand
- Solution to Poisson/Laplace eqn with given bc is unique
- and can be written down by simple superposition in this class of problems
- Basically a way to construct the Green's function by inspection

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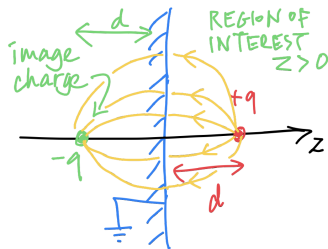
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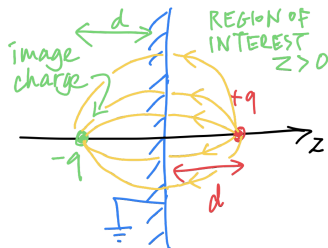
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- Simple example: point charge near grounded ∞ conducting plane
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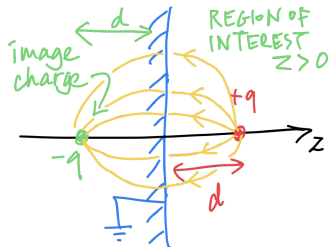
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- Solution

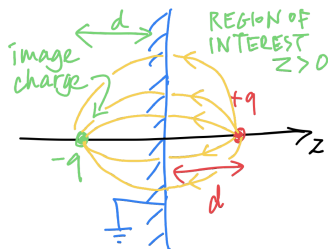


$$\Phi(\vec{x}) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{|\vec{x} - d\hat{z}|} - \frac{1}{|\vec{x} + d\hat{z}|} \right)$$

$$\Phi(\rho, z \geq 0) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + (z - d)^2}} - \frac{1}{\sqrt{\rho^2 + (z + d)^2}} \right)$$

Method of images

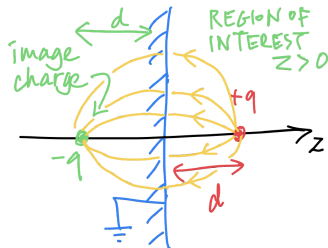
- Simple example: point charge near grounded ∞ conducting plane
- Image point charge (equal distance, opposite charge) behind plane (outside region) satisfies bc $\Phi(z=0) = 0$
- Math: Green's function



$$\begin{aligned} G_D(\vec{x}, \vec{x}') &= \frac{1}{|\vec{x} - \vec{x}'|} + F(\vec{x}, \vec{x}') = \frac{1}{|\vec{x} - \vec{x}'|} - \frac{1}{|\vec{x} - \vec{x}' + 2z'\hat{\mathbf{z}}|} \\ \rho(\vec{x}') &= q \delta(\vec{x}' - d\hat{\mathbf{z}}), \quad \nabla'^2 G = -4\pi\delta(\vec{x} - \vec{x}'), \quad \nabla'^2 F = 0, \quad z > 0 \\ \Phi(\vec{x}) &= \frac{1}{4\pi\epsilon_0} \int_V \rho(\vec{x}') G_D(\vec{x}, \vec{x}') d^3x' - \frac{1}{4\pi} \int_S \Phi(\vec{x}') \frac{\partial G_D(\vec{x}, \vec{x}')}{\partial n'} da' \\ &= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{|\vec{x} - d\hat{\mathbf{z}}|} - \frac{1}{|\vec{x} + d\hat{\mathbf{z}}|} \right) \end{aligned}$$

Method of images

- Simple example: point charge near grounded ∞ conducting plane
- Image point charge (equal distance, opposite charge) behind plane (outside region) satisfies bc $\Phi(z=0) = 0$
- Physics: charge distribution on conductor



$$\sigma(\rho) = \epsilon_0 \vec{E}(z=0) \cdot \hat{\mathbf{z}} = -\epsilon_0 \left. \frac{\partial \Phi}{\partial z} \right|_{z=0} = -\frac{q}{2\pi} \frac{d}{(d^2 + \rho^2)^{3/2}}$$

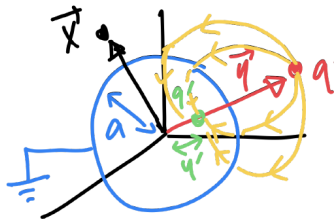
Exercise

Derive σ and show that $\int_S \sigma da = -q$

Grounded sphere

- Point charge q at location $\vec{y}$ near grounded conducting sphere with radius a centered at origin
- Image charge q' on line connecting origin to q with

$$q' = -\frac{a}{y} q, \quad y' = \frac{a^2}{y}$$

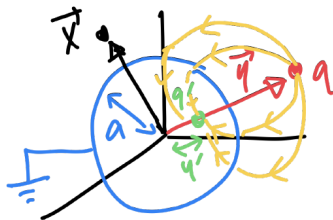


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- As the external charge approaches the surface ($y \rightarrow a$), so does the image ($y' \rightarrow a$) with $q' \rightarrow -q$

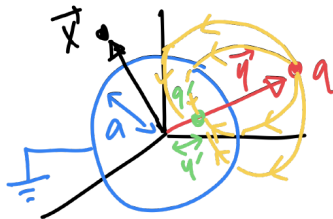


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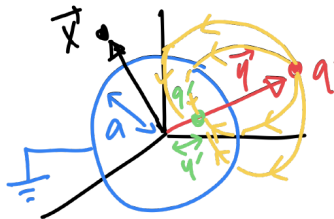
- As the charge moves away ($y \rightarrow \infty$), the image moves toward the origin ($y' \rightarrow 0$) with $q' \rightarrow 0$



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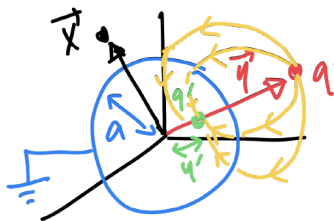
- Solution

$$\begin{aligned}\Phi(\vec{x}) &= \frac{1}{4\pi\epsilon_0} \left(\frac{q}{|\vec{x} - \vec{y}|} + \frac{q'}{|\vec{x} - \vec{y}'|} \right) \\ &= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{|\vec{x} - \vec{y}|} - \frac{a}{y} \frac{1}{\left| \vec{x} - \left(\frac{a}{y} \right)^2 \vec{y} \right|} \right]\end{aligned}$$

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- Induced surface charge density on the sphere

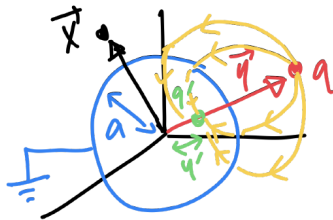
$$\sigma = -\epsilon_0 \left. \frac{\partial \Phi}{\partial x} \right|_{x=a} = -\frac{q}{4\pi a^2} \frac{\frac{a}{y} \left[1 - \left(\frac{a}{y} \right)^2 \right]}{\left[1 + \left(\frac{a}{y} \right)^2 - 2 \frac{a}{y} \cos \gamma \right]}, \quad \cos \gamma = \frac{\vec{x} \cdot \vec{y}}{x y}$$

symmetric around $\vec{y}$ and peaks at $\gamma = 0$

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- Force (attractive) from Coulomb's law for pt charges

$$|\vec{F}| = \frac{1}{4\pi\epsilon_0} \frac{q q'}{(y - y')^2} = \frac{1}{4\pi\epsilon_0} \frac{q \left(\frac{a}{y} q\right)}{\left(y - \frac{a^2}{y}\right)^2} = \frac{q^2}{4\pi\epsilon_0 y^2} \frac{\frac{a}{y}}{\left[1 - \left(\frac{a}{y}\right)^2\right]^2}$$

Charged insulated conducting sphere

- Charge Q on the sphere
- Use superposition: disconnect, add $Q - q'$ to sphere
- q' already at equilibrium, $Q - q'$ distributed uniformly, its contribution to potential equivalent to placing it at the origin

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$$\Phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|\vec{x} - \vec{y}|} - \frac{\left(\frac{a}{y}\right) q}{\left|\vec{x} - \left(\frac{a}{y}\right)^2 \vec{y}\right|} + \frac{Q + \left(\frac{a}{y}\right) q}{|\vec{x}|} \right]$$

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- Surface charge density

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- Use superposition: disconnect, add $Q - q'$ to sphere
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- Force on q

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \left[\frac{q q'}{(y - y')^2} + \frac{q(Q - q')}{y^2} \right] \frac{\vec{y}}{y}$$

Conducting sphere at fixed potential

- Sphere held at potential $V \neq 0$
- Same solution as charged sphere except charge at origin chosen as

$$\frac{Q_0}{4\pi\epsilon_0 a} = V$$

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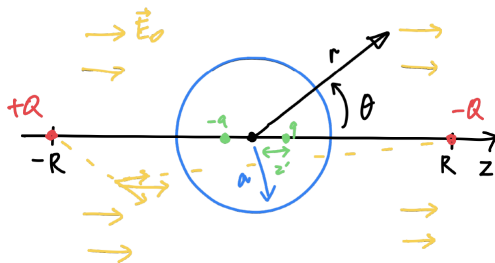
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Conducting sphere in uniform electric field

- $\vec{E} \rightarrow E_0 \hat{z}$ for $|z| \gg a$
- $\vec{E} \times \hat{r} = \vec{0}$ for $r = a$

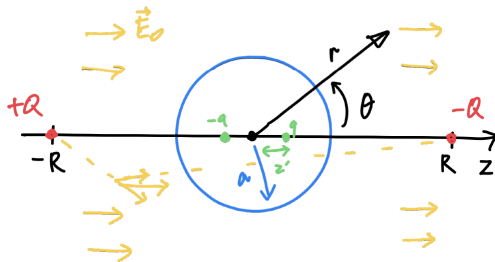


- Using image charges $\pm q$ at $\pm z'$ with

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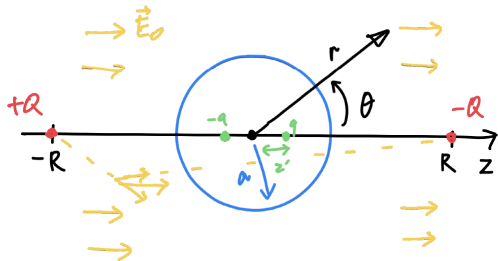


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- Generate $\vec{E}$ with charges $\pm Q$ at $z = \mp R$, $R \gg a$

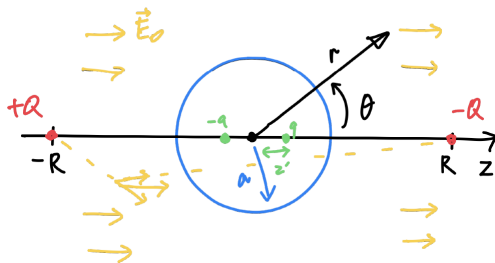
$$a \ll |z| \ll R, \quad \vec{E} \simeq \frac{2Q}{4\pi\epsilon_0 R^2} \hat{z} \rightarrow \frac{Q}{R^2} = 2\pi\epsilon_0 E_0$$

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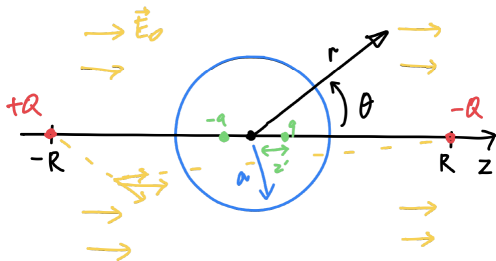


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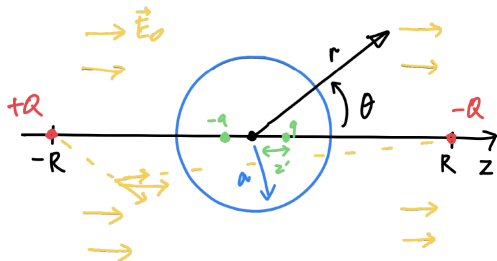
$$q = \frac{a}{R}Q, \quad z' = \frac{a^2}{R}$$

- the solution is the $R \rightarrow \infty$ limit (at fixed E_0) of

$$\Phi = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{|\vec{r} + R\hat{z}|} + \frac{-Q}{|\vec{r} - R\hat{z}|} + \frac{q}{|\vec{r} - z'\hat{z}|} + \frac{-q}{|\vec{r} + z'\hat{z}|} \right)$$

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$$\Phi(r, \theta) = -E_0 r \left[1 - \left(\frac{a}{r} \right)^3 \right] \cos \theta$$

What we're really after

- Once we find the Green function $G_D(\vec{x}, \vec{x}')$ for a given geometry (using the method of images, for example), we can use the general form

$$\Phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int_V \rho(\vec{x}') G_D(\vec{x}, \vec{x}') d^3x' - \frac{1}{4\pi} \int_S \Phi(\vec{x}') \frac{\partial G_D(\vec{x}, \vec{x}')}{\partial n'} da'$$

- The solution to Laplace's eqn ($\rho(\vec{x}') = 0$) for the Dirichlet BVP (Φ specified on S) with that geometry is

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Green function for the sphere

- Substitute $\vec{y} \rightarrow \vec{x}'$ for the source point

$$\begin{aligned} G(\vec{x}, \vec{x}') &= \frac{1}{|\vec{x} - \vec{x}'|} - \frac{a}{x' \left| \vec{x} - \left(\frac{a}{x'}\right)^2 \vec{x}' \right|} \\ &= \frac{1}{\sqrt{(x^2 + x'^2 - 2xx' \cos \gamma)}} - \frac{1}{\sqrt{\left(\frac{xx'}{a}\right)^2 + a^2 - 2xx' \cos \gamma}} \end{aligned}$$

- which satisfies

$$G(\vec{x}', \vec{x}) = G(\vec{x}, \vec{x}'); \quad G(x = a, \vec{x}') = G(\vec{x}, x' = a) = 0$$

- Also need normal derivative on the surface ($\hat{n}' = \vec{x}'/x'$)

$$\left. \frac{\partial G}{\partial n'} \right|_{x'=a} = - \frac{x^2 - a^2}{a (x^2 + a^2 - 2ax \cos \gamma)^{3/2}}$$

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Green function for the sphere

- General solution to Laplace's eqn outside the sphere for *any* BVP with Φ given on the sphere

$$\Phi(\vec{x}) = \frac{1}{4\pi} \int \Phi(a, \theta', \phi') \frac{a(x^2 - a^2)}{(x^2 + a^2 - 2ax \cos \gamma)^{3/2}} d\Omega'$$

- where γ is the angle between $\vec{x}$ and $\vec{x}'$

$$\vec{x} = (a \sin \theta \cos \phi, a \sin \theta \sin \phi, a \cos \theta)$$

$$\vec{x}' = (a \sin \theta' \cos \phi', a \sin \theta' \sin \phi', a \cos \theta')$$

$$\cos \gamma = \frac{\vec{x} \cdot \vec{x}'}{a^2} = \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\phi - \phi')$$

- Sign reversed (surface normal) for solution inside the sphere

Green function for the sphere

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Conducting sphere, upper/lower hemisphere at $\pm V$

- Using Green fn

$$\begin{aligned}\Phi(\vec{x}) &= \frac{1}{4\pi} \int \Phi(a, \theta', \phi') \frac{a(x^2 - a^2)}{(x^2 + a^2 - 2ax \cos \gamma)^{3/2}} d\Omega' \\ &= \frac{Va}{4\pi} (x^2 - a^2) \int_0^{2\pi} d\phi' \left[\int_0^1 \frac{d \cos \theta'}{(x^2 + a^2 - 2ax \cos \gamma)^{3/2}} \right. \\ &\quad \left. - \int_{-1}^0 \frac{d \cos \theta'}{(x^2 + a^2 - 2ax \cos \gamma)^{3/2}} \right]\end{aligned}$$

Conducting sphere, upper/lower hemisphere at $\pm V$

- Using Green fn

$$\begin{aligned}\Phi(\vec{x}) &= \frac{1}{4\pi} \int \Phi(a, \theta', \phi') \frac{a(x^2 - a^2)}{(x^2 + a^2 - 2ax \cos \gamma)^{3/2}} d\Omega' \\ &= \frac{Va}{4\pi} (x^2 - a^2) \int_0^{2\pi} d\phi' \left[\int_0^1 \frac{d \cos \theta'}{(x^2 + a^2 - 2ax \cos \gamma)^{3/2}} \right. \\ &\quad \left. - \int_{-1}^0 \frac{d \cos \theta'}{(x^2 + a^2 - 2ax \cos \gamma)^{3/2}} \right]\end{aligned}$$

- Substitution in 2nd term

$$\cos \gamma = \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\phi - \phi')$$

$$\theta' \rightarrow \pi - \theta', \quad \cos \theta' \rightarrow -\cos \theta', \quad \sin \theta' \rightarrow \sin \theta'$$

$$\phi' \rightarrow \pi + \phi', \quad \cos(\phi - \phi') \rightarrow -\cos(\phi - \phi')$$

$$\cos \gamma \rightarrow -\cos \gamma$$

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$$\phi' \rightarrow \pi + \phi', \cos(\phi - \phi') \rightarrow -\cos(\phi - \phi')$$

$$\cos \gamma \rightarrow -\cos \gamma$$

- gives

$$\Phi(\vec{x}) = \frac{Va}{4\pi}(x^2 - a^2) \int_0^{2\pi} d\phi' \int_0^1 d\cos \theta' \left[\frac{1}{(x^2 + a^2 - 2ax \cos \gamma)^{3/2}} - \frac{1}{(x^2 + a^2 + 2ax \cos \gamma)^{3/2}} \right]$$

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- gives

$$\Phi(\vec{x}) = \frac{Va}{4\pi}(x^2 - a^2) \int_0^{2\pi} d\phi' \int_0^1 d\cos\theta' \left[\frac{1}{(x^2 + a^2 - 2ax\cos\gamma)^{3/2}} - \frac{1}{(x^2 + a^2 + 2ax\cos\gamma)^{3/2}} \right]$$

- which, for the +z axis, simplifies to

$$\theta = 0, \cos\gamma = \cos\theta', x = z$$

$$\Phi(z) = \frac{Va}{4\pi}(z^2 - a^2) \int_0^{2\pi} d\phi' \int_0^1 d\cos\theta' \left[\frac{1}{(z^2 + a^2 - 2az\cos\theta')^{3/2}} - \frac{1}{(z^2 + a^2 + 2az\cos\theta')^{3/2}} \right]$$

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$$\theta = 0, \cos \gamma = \cos \theta', x = z$$

$$\begin{aligned}\Phi(z) &= \frac{V a}{4\pi} (z^2 - a^2) \int_0^{2\pi} d\phi' \int_0^1 d\cos \theta' \\ &\quad \left[\frac{1}{(z^2 + a^2 - 2aZ \cos \theta')^{3/2}} - \frac{1}{(z^2 + a^2 + 2aZ \cos \theta')^{3/2}} \right] \\ &= \frac{V a}{4\pi} (z^2 - a^2) (2\pi) \frac{1}{2aZ} \int_0^1 du \\ &\quad \left[(z^2 + a^2 - u)^{-3/2} - (z^2 + a^2 + u)^{-3/2} \right]\end{aligned}$$

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$$\theta = 0, \cos \gamma = \cos \theta', x = z$$

$$\begin{aligned}\Phi(z) &= \frac{V a}{4\pi} (z^2 - a^2) \int_0^{2\pi} d\phi' \int_0^1 d\cos \theta' \\ &\quad \left[\frac{1}{(z^2 + a^2 - 2aZ \cos \theta')^{3/2}} - \frac{1}{(z^2 + a^2 + 2aZ \cos \theta')^{3/2}} \right] \\ &= \frac{V a}{4\pi} (z^2 - a^2) (2\pi) \frac{1}{2aZ} \int_0^1 du \\ &\quad \left[(z^2 + a^2 - u)^{-3/2} - (z^2 + a^2 + u)^{-3/2} \right] \\ &= \frac{V}{4} \frac{z^2 - a^2}{z} (-2) \left[(z^2 + a^2 - u)^{-1/2} - (z^2 + a^2 + u)^{-1/2} \right]_0^1\end{aligned}$$

Conducting sphere, upper/lower hemisphere at $\pm V$

$$\theta = 0, \cos \gamma = \cos \theta', x = z$$

$$\Phi(z) = \frac{V a}{4\pi} (z^2 - a^2) \int_0^{2\pi} d\phi' \int_0^1 d\cos \theta'$$

$$\left[\frac{1}{(z^2 + a^2 - 2az \cos \theta')^{3/2}} - \frac{1}{(z^2 + a^2 + 2az \cos \theta')^{3/2}} \right]$$

$$= \frac{V a}{4\pi} (z^2 - a^2) (2\pi) \frac{1}{2az} \int_0^1 du$$

$$\left[(z^2 + a^2 - u)^{-3/2} - (z^2 + a^2 + u)^{-3/2} \right]$$

$$= \frac{V}{4} \frac{z^2 - a^2}{z} (-2) \left[(z^2 + a^2 - u)^{-1/2} - (z^2 + a^2 + u)^{-1/2} \right]_0^1$$

$$= \frac{V}{2} \frac{z^2 - a^2}{z} \left[\frac{1}{\sqrt{z^2 + a^2 + 2az}} - \frac{2}{\sqrt{z^2 + a^2}} + \frac{1}{\sqrt{z^2 + a^2 - 2az}} \right]$$

Conducting sphere, upper/lower hemisphere at $\pm V$

$$\begin{aligned} &= \frac{V a}{4\pi} (z^2 - a^2) (2\pi) \frac{1}{2az} \int_0^1 du \\ &\quad \left[\left(z^2 + a^2 - u \right)^{-3/2} - \left(z^2 + a^2 + u \right)^{-3/2} \right] \\ &= \frac{V}{4} \frac{z^2 - a^2}{z} (-2) \left[\left(z^2 + a^2 - u \right)^{-1/2} - \left(z^2 + a^2 + u \right)^{-1/2} \right]_0^1 \\ &= \frac{V}{2} \frac{z^2 - a^2}{z} \left[\frac{1}{\sqrt{z^2 + a^2 + 2az}} - \frac{2}{\sqrt{z^2 + a^2}} + \frac{1}{\sqrt{z^2 + a^2 - 2az}} \right] \\ &= \frac{V}{2} \frac{(z - a)(z + a)}{z} \left[\frac{1}{z + a} - \frac{2}{\sqrt{z^2 + a^2}} + \frac{1}{|z - a|} \right] \end{aligned}$$

Conducting sphere, upper/lower hemisphere at $\pm V$

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$$\begin{aligned} &= \frac{V}{4} \frac{z^2 - a^2}{z} (-2) \left[\left(z^2 + a^2 - u \right)^{-1/2} - \left(z^2 + a^2 + u \right)^{-1/2} \right]_0^1 \\ &= \frac{V}{2} \frac{z^2 - a^2}{z} \left[\frac{1}{\sqrt{z^2 + a^2 + 2az}} - \frac{2}{\sqrt{z^2 + a^2}} + \frac{1}{\sqrt{z^2 + a^2 - 2az}} \right] \\ &= \frac{V}{2} \frac{(z - a)(z + a)}{z} \left[\frac{1}{z + a} - \frac{2}{\sqrt{z^2 + a^2}} + \frac{1}{|z - a|} \right] \\ &= \frac{V}{2z} \left[(z - a) - \frac{2(z^2 - a^2)}{\sqrt{z^2 + a^2}} + (z + a) \right] \\ \Phi(z > a) &= V \left[1 - \frac{z^2 - a^2}{z\sqrt{z^2 + a^2}} \right] \end{aligned}$$

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$$\begin{aligned} &= \frac{V}{4} \frac{z^2 - a^2}{z} (-2) \left[\left(z^2 + a^2 - u \right)^{-1/2} - \left(z^2 + a^2 + u \right)^{-1/2} \right]_0^1 \\ &= \frac{V}{2} \frac{z^2 - a^2}{z} \left[\frac{1}{\sqrt{z^2 + a^2 + 2az}} - \frac{2}{\sqrt{z^2 + a^2}} + \frac{1}{\sqrt{z^2 + a^2 - 2az}} \right] \\ &= \frac{V}{2} \frac{(z - a)(z + a)}{z} \left[\frac{1}{z + a} - \frac{2}{\sqrt{z^2 + a^2}} + \frac{1}{|z - a|} \right] \\ &= \frac{V}{2z} \left[(z - a) - \frac{2(z^2 - a^2)}{\sqrt{z^2 + a^2}} + (z + a) \right] \\ \Phi(z > a) &= V \left[1 - \frac{z^2 - a^2}{z\sqrt{z^2 + a^2}} \right] \\ \Phi(z < -a) &= -\Phi(|z|) \end{aligned}$$

Conducting sphere, upper/lower hemisphere at $\pm V$

$$\Phi(z > a) = V \left[1 - \frac{z^2 - a^2}{z\sqrt{z^2 + a^2}} \right]$$

- $\Phi \rightarrow V$ as $z \rightarrow a$
- Asymptotic behavior ($z \gg a$)

$$\begin{aligned}\alpha &= \frac{a}{z} \ll 1 \\ \Phi(z \gg a) &= V \left[1 - \frac{1 - \alpha^2}{\sqrt{1 + \alpha^2}} \right] \\ &= \left\{ 1 - (1 - \alpha^2) \left[1 - \frac{1}{2}\alpha^2 + \mathcal{O}(\alpha^4) \right] \right\} \simeq \frac{3}{2} V \left(\frac{a}{z} \right)^2\end{aligned}$$

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Orthogonal functions

- Set of functions $\{U_n(\xi)\}$ square integrable over an interval (a, b) called *orthogonal* if

$$\int_a^b U_n^*(\xi) U_m(\xi) d\xi = 0, m \neq n$$

- and *orthonormal* if the integrals are normalized to 1

$$\int_a^b U_n^*(\xi) U_m(\xi) d\xi = \delta_{nm} = \begin{cases} 0, & m \neq n \\ 1, & m = n \end{cases}$$

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Expansion in orthonormal functions

- For a square integrable function $f(\xi)$ on (a, b) , we can form the sum

$$f_N(\xi) = \sum_{n=1}^N c_n U_n(\xi)$$

- and compute the “best” fit for the coefficients c_n to minimize the *mean square error* (MSE) M_N

$$M_N = \int_a^b \left| f(\xi) - \sum_{n=1}^N c_n U_n(\xi) \right|^2 d\xi$$

- which yields

$$\frac{\partial M_N}{\partial c_n^*} = 0 \rightarrow c_n = \int_a^b U_n^*(\xi) f(\xi) d\xi$$

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Expansion in orthonormal functions

- Same thing in Dirac notation

$$U_n \leftrightarrow |U_n\rangle$$

$$\langle f|g\rangle = \int_a^b f^*(\xi)g(\xi) d\xi$$

$$\langle U_n|U_m\rangle = \delta_{nm}$$

$$|f_N\rangle = \sum_n c_n |U_n\rangle$$

$$c_n = \langle U_n|f\rangle$$

Complete sets of functions

- For a *complete* set of functions $\{U_n(\xi)\}$, the expansion converges to the function as the number of terms is increased

$$\lim_{N \rightarrow \infty} f_N(\xi) = f(\xi) = \sum_{n=1}^{\infty} c_n U_n(\xi)$$

- also expressed as the *closure relation*

$$\begin{aligned} f(\xi) &= \sum_{n=1}^{\infty} c_n U_n(\xi) = \sum_{n=1}^{\infty} \left[\int_a^b U_n^*(\xi') f(\xi') d\xi' \right] U_n(\xi) \\ &= \int_a^b \left[\sum_{n=1}^{\infty} U_n^*(\xi') U_n(\xi) \right] f(\xi') d\xi' \\ &\rightarrow \sum_{n=1}^{\infty} U_n^*(\xi') U_n(\xi) = \delta(\xi' - \xi) \end{aligned}$$

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Fourier series

- Sines and cosines as orthonormal basis functions in the interval $(-a/2, a/2)$

$$\sqrt{\frac{2}{a}} \sin\left(m \frac{2\pi}{a} x\right), \quad \sqrt{\frac{2}{a}} \cos\left(m \frac{2\pi}{a} x\right); \quad m > 0$$

- with the series usually written as

$$f(x) = \frac{1}{2}A_0 + \sum_{m=1}^{\infty} \left[A_m \cos\left(m \frac{2\pi}{a} x\right) + B_m \sin\left(m \frac{2\pi}{a} x\right) \right]$$

$$A_m = \frac{2}{a} \int_{-a/2}^{a/2} \cos\left(m \frac{2\pi}{a} x\right) f(x) dx$$

$$B_m = \frac{2}{a} \int_{-a/2}^{a/2} \sin\left(m \frac{2\pi}{a} x\right) f(x) dx$$

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- Generalization to 2D interval $(a, b) \times (c, d)$ with two sets of complete functions $\{U_n\}$, $\{V_m\}$

$$f(\xi, \eta) = \sum_n \sum_m c_{nm} U_n(\xi) V_m(\eta)$$

$$c_{nm} = \int_a^b d\xi \int_c^d d\eta U_n^*(\xi) V_m^*(\eta) f(\xi, \eta)$$

Continuous index

- The functions $\{U_n\}$ can also be defined on an infinite interval and form an uncountable set
- Fourier series $\rightarrow$ Fourier integral (transform)

Continuous index

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$$x \in (-a/2, a/2) \quad m = 0, \pm 1, \pm 2, \dots$$

$$U_m(x) = \frac{1}{\sqrt{a}} e^{i(2\pi mx/a)}$$

$$\delta_{nm} = \int_{-a/2}^{a/2} U_m^*(x) U_n(x) dx$$

$$f(x) = \sum_m A_m U_m(x)$$

$$= \frac{1}{\sqrt{a}} \sum_{m=-\infty}^{\infty} A_m e^{i(2\pi mx/a)}$$

$$A_m = \frac{1}{\sqrt{a}} \int_{-a/2}^{a/2} e^{-i(2\pi mx/a)} f(x) dx$$

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$$A_m = \frac{1}{\sqrt{a}} \int_{-a/2}^{a/2} e^{-i(2\pi mx/a)} f(x) dx$$

$$a \rightarrow \infty, \quad 2\pi m/a \rightarrow k$$

$$U_k(x) = \frac{1}{\sqrt{2\pi}} e^{ikx}$$

$$\delta(k - k') = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(k-k')x} dx$$

$$f(x) = \int_{-\infty}^{\infty} A(k) U_k(x) dk$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx} f(x) dx$$

$$A(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} f(x) dx$$

Separation of variables

- Laplace equation in rectangular coordinates

$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2} = 0$$

- Assuming $\Phi(x, y, z) = X(x)Y(y)Z(z)$,

$$\frac{1}{\Phi} \nabla^2 \Phi = \frac{1}{X(x)} \frac{d^2 X}{dx^2} + \frac{1}{Y(y)} \frac{d^2 Y}{dy^2} + \frac{1}{Z(z)} \frac{d^2 Z}{dz^2} = 0$$

- which holds if each term is constant, eg. $\alpha^2 + \beta^2 = \gamma^2$

$$\frac{1}{X(x)} \frac{d^2 X}{dx^2} = -\alpha^2$$

$$\frac{1}{Y(y)} \frac{d^2 Y}{dy^2} = -\beta^2$$

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Laplace equation in rectangular coordinates

- With this particular (arbitrary) choice ($\alpha^2, \beta^2, \gamma^2 > 0$)

$$\frac{1}{X(x)} \frac{d^2 X}{dx^2} = -\alpha^2 \rightarrow X(x) \propto e^{\pm i\alpha x}$$

$$\frac{1}{Y(y)} \frac{d^2 Y}{dy^2} = -\beta^2 \rightarrow Y(y) \propto e^{\pm i\beta y}$$

$$\frac{1}{Z(z)} \frac{d^2 Z}{dz^2} = \gamma^2 \rightarrow Z(z) \propto e^{\pm \gamma z}$$

$$\Phi(x, y, z) = X(x)Y(y)Z(z) \propto e^{\pm i\alpha x} e^{\pm i\beta y} e^{\pm \sqrt{\alpha^2 + \beta^2} z}$$

- with the details to be determined by boundary conditions

Laplace equation in rectangular coordinates

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$$\frac{1}{X(x)} \frac{d^2 X}{dx^2} = -\alpha^2 \rightarrow X(x) \propto e^{\pm i\alpha x}$$

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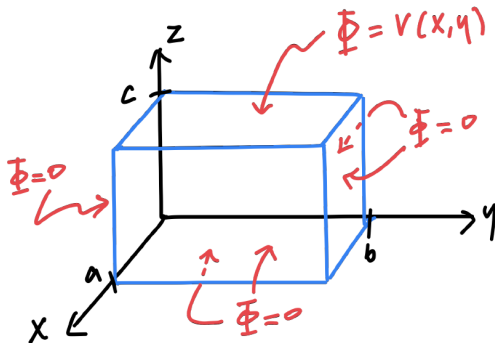
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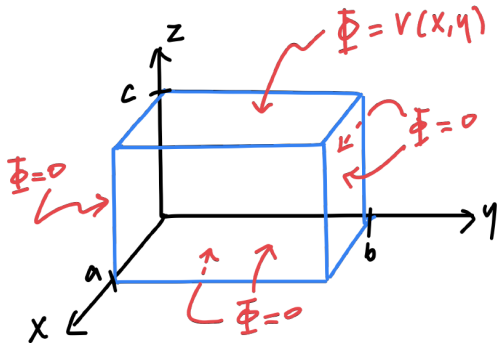
Laplace equation in rectangular coordinates

- Rectangular box with dimensions a, b, c
- bc $\Phi = 0$ on all surfaces except $z = c$



Laplace equation in rectangular coordinates

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- From separation of variables

$$X \propto \sin(\alpha_n x), \quad \alpha_n a = n\pi$$

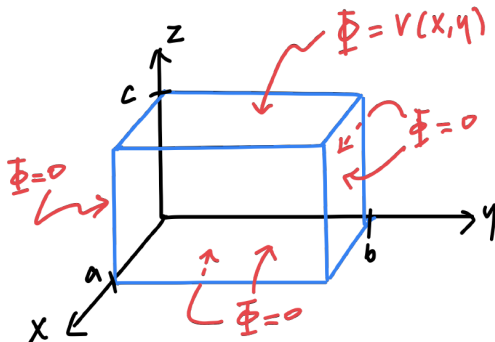
$$Y \propto \sin(\beta_m y), \quad \beta_m b = m\pi$$

$$Z \propto \sinh(\gamma_{nm} z), \quad \gamma_{nm} = \pi \sqrt{\left(\frac{n}{a}\right)^2 + \left(\frac{m}{b}\right)^2}$$

$$\Phi_{nm}(x, y, z) = \sin(\alpha_n x) \sin(\beta_m y) \sinh(\gamma_{nm} z)$$

Laplace equation in rectangular coordinates

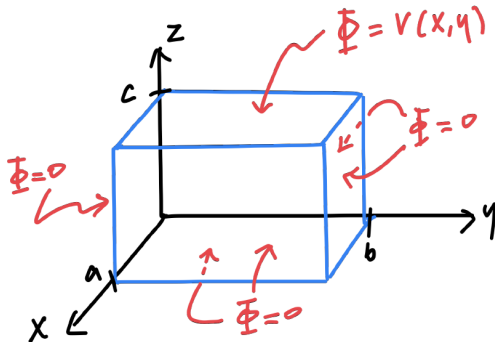
- Rectangular box with dimensions a, b, c
- bc $\Phi = 0$ on all surfaces except $z = c$
- Solution



$$\Phi(x, y, z) = \sum_{n,m=1}^{\infty} A_{nm} \sin(\alpha_n x) \sin(\beta_m y) \sinh(\gamma_{nm} z)$$

Laplace equation in rectangular coordinates

- Rectangular box with dimensions a, b, c
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- Solution

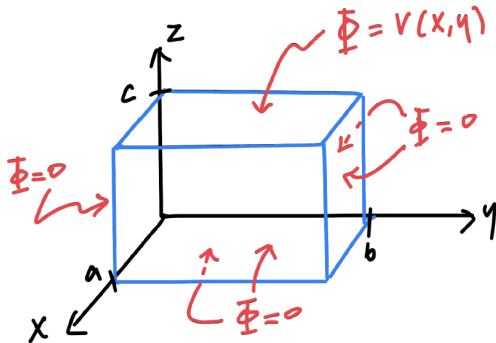
$$\Phi(x, y, z) = \sum_{n,m=1}^{\infty} A_{nm} \sin(\alpha_n x) \sin(\beta_m y) \sinh(\gamma_{nm} z)$$

- with the coefficients determined by remaining bc at $z = c$

$$\Phi(z = c) = V(x, y) = \sum_{n,m=1}^{\infty} \sinh(\gamma_{nm} c) A_{nm} \sin(\alpha_n x) \sin(\beta_m y)$$

Laplace equation in rectangular coordinates

- Rectangular box with dimensions a, b, c
- bc $\Phi = 0$ on all surfaces except $z = c$



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$$\Phi(z = c) = V(x, y) = \sum_{n,m=1}^{\infty} \sinh(\gamma_{nm}c) A_{nm} \sin(\alpha_n x) \sin(\beta_m y)$$

- 2D Fourier series for $V(x, y)$

$$A_{nm} = \frac{4}{ab \sinh(\gamma_{nm}c)} \int_0^a dx \int_0^b dy V(x, y) \sin(\alpha_n x) \sin(\beta_m y)$$

2D example

- If $\partial\Phi/\partial z = 0$ due to symmetry (eg. very long structure with uniform cross section)

$$\Phi(x, y) = X(x)Y(y)$$

$$\frac{1}{\Phi} \nabla^2 \Phi = \frac{1}{X(x)} \frac{d^2 X}{dx^2} + \frac{1}{Y(y)} \frac{d^2 Y}{dy^2} = 0$$

$$\frac{1}{X(x)} \frac{d^2 X}{dx^2} = -\frac{1}{Y(y)} \frac{d^2 Y}{dy^2} = -\alpha^2$$

$$\Phi(x, y) \propto e^{\pm i\alpha x} e^{\pm \alpha y}$$

- allowable values of α and coefficients in the superposition determined by bc

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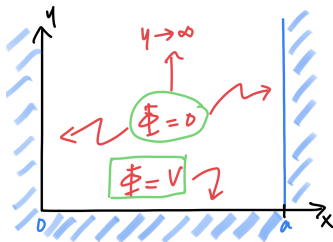
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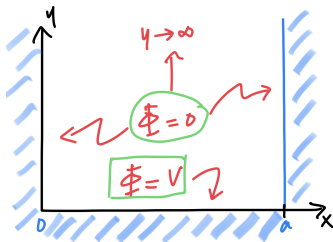
2D example

- semi- ∞ box in x - y plane
- with $\Phi = 0$ at $x = 0, a$ and $y \rightarrow \infty$
- want sin for $\Phi = 0$ at $x = 0, a$
- and decaying exponential for $\Phi \rightarrow 0$ at $y \rightarrow \infty$



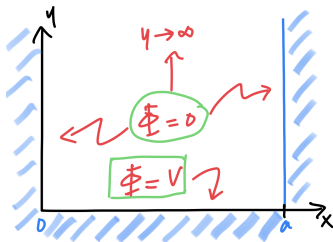
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2D example

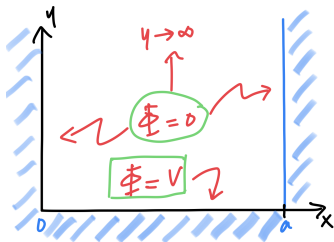
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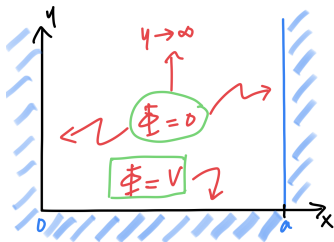
$$\Phi(x, y) = \sum_{n=1}^{\infty} A_n \sin(n\pi x/a) \exp(-n\pi y/a)$$



2D example

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$$\Phi(x, y) = \sum_{n=1}^{\infty} A_n \sin(n\pi x/a) \exp(-n\pi y/a)$$



- bc at $y = 0$ gives another Fourier series

$$\Phi(x, 0) = \sum_{n=1}^{\infty} A_n \sin(n\pi x/a)$$

$$A_n = \frac{2}{a} \int_0^a \Phi(x, 0) \sin(n\pi x/a) dx$$

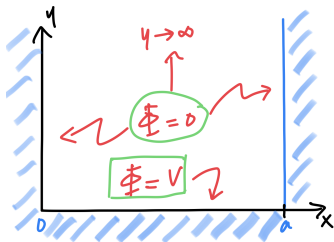
2D example

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- If $\Phi(x, 0) = V$

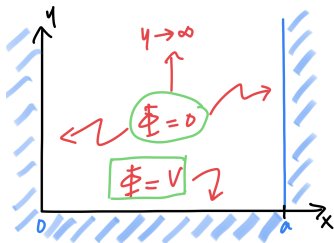
$$A_n = \begin{cases} \frac{4V}{n\pi}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$



2D example

- semi- ∞ box in x - y plane
- with $\Phi = 0$ at $x = 0, a$ and $y \rightarrow \infty$
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- If $\Phi(x, 0) = V$

$$A_n = \begin{cases} \frac{4V}{n\pi}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$

- and the solution becomes

$$\Phi(x, y) = \frac{4V}{\pi} \sum_{n \text{ odd}} \frac{1}{n} \exp\left(-n\frac{\pi}{a}y\right) \sin\left(n\frac{\pi}{a}x\right)$$

2D example

- Solution

$$\begin{aligned}\Phi(x, y) &= \frac{4V}{\pi} \sum_{n=0}^{\infty} \frac{1}{2n+1} \exp \left[-(2n+1) \frac{\pi}{a} y \right] \sin \left[(2n+1) \frac{\pi}{a} x \right] \\ &= \frac{4V}{\pi} \exp \left(-\frac{\pi}{a} y \right) \left\{ \sin \left(\frac{\pi}{a} x \right) + \frac{1}{3} \left[\exp \left(-\frac{\pi}{a} y \right) \right]^2 \sin \left(\frac{3\pi}{a} x \right) \right. \\ &\quad \left. + \frac{1}{5} \left[\exp \left(-\frac{\pi}{a} y \right) \right]^4 \sin \left(\frac{5\pi}{a} x \right) + \dots \right\}\end{aligned}$$

- decays as

$$\Phi(x, y) \rightarrow \frac{4V}{\pi} \exp \left(-\frac{\pi}{a} y \right) \sin \left(\frac{\pi}{a} x \right), \quad y \gtrsim a$$

2D example

- Solution

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2D example

- Series can be summed for exact solution in this case

$$X + iY = \frac{\pi}{a}(x + iy), \quad Z = e^{i(X+iY)} = (\cos X + i \sin X) e^{-Y}$$

$$\operatorname{Im} \left\{ e^{in(X+iY)} \right\} = \sin nX e^{-nY}, \quad \Phi(x, y) = \frac{4V}{\pi} \operatorname{Im} \left\{ \sum_{n \text{ odd}} \frac{Z^n}{n} \right\}$$

$$\sum_{n \text{ odd}} \frac{Z^n}{n} = \frac{1}{2} \ln \left(\frac{1+Z}{1-Z} \right), \quad \frac{1+Z}{1-Z} = \frac{1 - |Z|^2 + 2i \operatorname{Im}\{Z\}}{1 - |Z|^2}$$

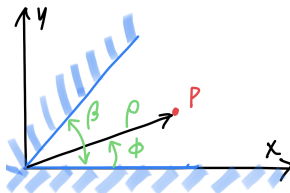
$$z = |z|e^{i\theta}, \quad \operatorname{Im}\{\ln z\} = \theta, \quad \tan \theta = \frac{2 \operatorname{Im}\{Z\}}{1 - |Z|^2} = \frac{2 \sin X e^{-Y}}{1 - e^{-2Y}} = \frac{\sin X}{\sinh Y}$$

$$\Phi(x, y) = \frac{2V}{\pi} \tan^{-1} \left[\frac{\sin \left(\frac{\pi}{a} x \right)}{\sinh \left(\frac{\pi}{a} y \right)} \right]$$

Field at corners and edges in 2D

- Conducting planes meet at angle β
- Near the corner (origin),

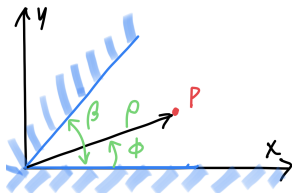
$$\nabla^2 \Phi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \Phi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \Phi}{\partial \phi^2} = 0$$



Field at corners and edges in 2D

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- Near the corner (origin),

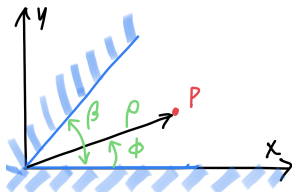
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Field at corners and edges in 2D

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$$\nabla^2 \Phi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \Phi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \Phi}{\partial \phi^2} = 0$$



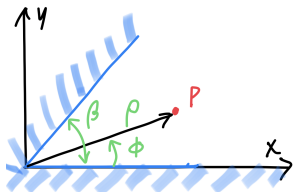
- Separation of variables gives

$$\Phi(\rho, \phi) = R(\rho) \Psi(\phi), \quad \frac{\rho^2}{\Phi} \nabla^2 \Phi = \frac{\rho}{R} \frac{d}{d\rho} \left(\rho \frac{dR}{d\rho} \right) + \frac{1}{\Psi} \frac{d^2 \Psi}{d\phi^2} = 0$$

Field at corners and edges in 2D

- Conducting planes meet at angle β
- Near the corner (origin),

$$\nabla^2 \Phi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \Phi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \Phi}{\partial \phi^2} = 0$$



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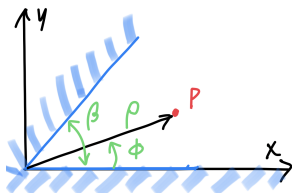
- leading to

$$\frac{\rho}{R} \frac{d}{d\rho} \left(\rho \frac{dR}{d\rho} \right) = \nu^2, \quad \frac{1}{\Psi} \frac{d^2 \Psi}{d\phi^2} = -\nu^2$$

Field at corners and edges in 2D

- Conducting planes meet at angle β
- Near the corner (origin),

$$\nabla^2 \Phi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \Phi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \Phi}{\partial \phi^2} = 0$$



- leading to

$$\frac{\rho}{R} \frac{d}{d\rho} \left(\rho \frac{dR}{d\rho} \right) = \nu^2, \quad \frac{1}{\psi} \frac{d^2 \psi}{d\phi^2} = -\nu^2$$

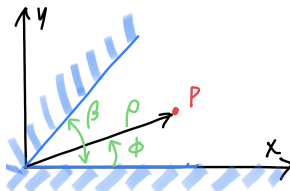
- with solutions, for $\nu \neq 0$

$$R(\rho) = a\rho^\nu + b\rho^{-\nu}, \quad \Psi(\phi) = A \cos(\nu\phi) + B \sin(\nu\phi)$$

Field at corners and edges in 2D

- Conducting planes meet at angle β
- Near the corner (origin),

$$\nabla^2 \Phi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \Phi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \Phi}{\partial \phi^2} = 0$$



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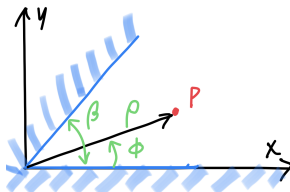
- and, for $\nu = 0$

$$R(\rho) = a_0 + b_0 \ln \rho, \quad \Psi(\phi) = A_0 + B_0 \phi$$

Field at corners and edges in 2D

- Conducting planes meet at angle β
- Near the corner (origin),

$$\nabla^2 \Phi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \Phi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \Phi}{\partial \phi^2} = 0$$



- with solutions, for $\nu \neq 0$

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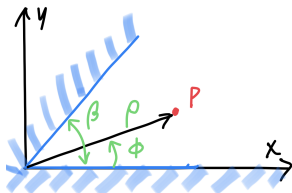
- and from the bc

$$\Phi(\rho, \beta) = \Phi(\rho, 0) = \Phi(0, \phi) \rightarrow B_0 = 0, \quad b_0 = 0$$

Field at corners and edges in 2D

- Conducting planes meet at angle β
- Near the corner (origin),

$$\nabla^2 \Phi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \Phi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \Phi}{\partial \phi^2} = 0$$



- with solutions, for $\nu \neq 0$

$$R(\rho) = a\rho^\nu + b\rho^{-\nu}, \quad \Psi(\phi) = A\cos(\nu\phi) + B\sin(\nu\phi)$$

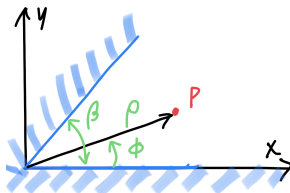
- For $\Phi = V$ on conductors,

$$\nu\beta = m\pi, \quad \Phi(\rho, \phi) = V + \sum_{m=1}^{\infty} a_m \rho^{m\pi/\beta} \sin(m\pi\phi/\beta)$$

Field at corners and edges in 2D

- Conducting planes meet at angle β
- Near the corner (origin),

$$\nabla^2 \Phi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \Phi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \Phi}{\partial \phi^2} = 0$$



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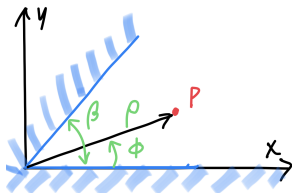
- and near the origin

$$\Phi(\rho, \phi) \simeq V + a_1 \rho^{\pi/\beta} \sin(\pi\phi/\beta)$$

Field at corners and edges in 2D

- Conducting planes meet at angle β
- Near the corner (origin),

$$\nabla^2 \Phi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \Phi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \Phi}{\partial \phi^2} = 0$$



- and near the origin

$$\Phi(\rho, \phi) \simeq V + a_1 \rho^{\pi/\beta} \sin(\pi\phi/\beta)$$

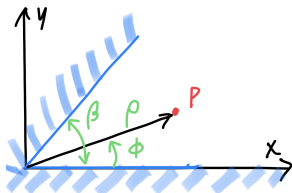
$$E_\rho = -\frac{\partial \Phi}{\partial \rho} \simeq -\frac{\pi a_1}{\beta} \rho^{\pi/\beta-1} \sin(\pi\phi/\beta)$$

$$E_\phi = -\frac{1}{\rho} \frac{\partial \Phi}{\partial \phi} \simeq -\frac{\pi a_1}{\beta} \rho^{\pi/\beta-1} \cos(\pi\phi/\beta)$$

Field at corners and edges in 2D

- Conducting planes meet at angle β
- Near the corner (origin),

$$\nabla^2 \Phi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \Phi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \Phi}{\partial \phi^2} = 0$$



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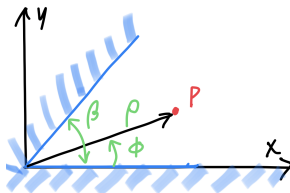
$$\sigma(\rho) = \epsilon_0 E_\phi(\rho, 0) \simeq -\frac{\epsilon_0 \pi a_1}{\beta} \rho^{\pi/\beta-1} \cos(\pi\phi/\beta)$$

Field at corners and edges in 2D

$$E_\rho \propto \frac{1}{\beta} \rho^{\pi/\beta-1} \sin(\pi\phi/\beta)$$

$$E_\phi \propto \frac{1}{\beta} \rho^{\pi/\beta-1} \cos(\pi\phi/\beta)$$

$$\sigma(\rho) \propto \frac{1}{\beta} \rho^{\pi/\beta-1} \cos(\pi\phi/\beta)$$

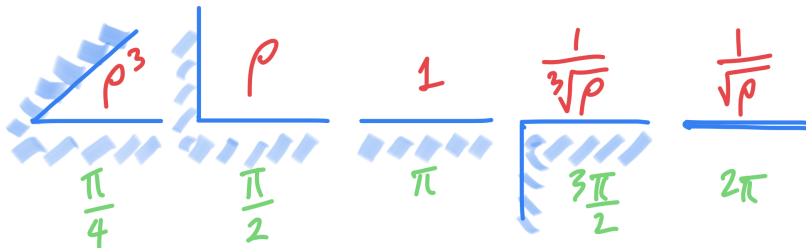
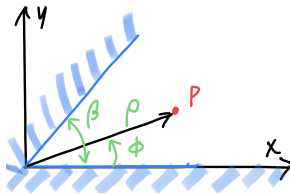


Field at corners and edges in 2D

$$E_\rho \propto \frac{1}{\beta} \rho^{\pi/\beta-1} \sin(\pi\phi/\beta)$$

$$E_\phi \propto \frac{1}{\beta} \rho^{\pi/\beta-1} \cos(\pi\phi/\beta)$$

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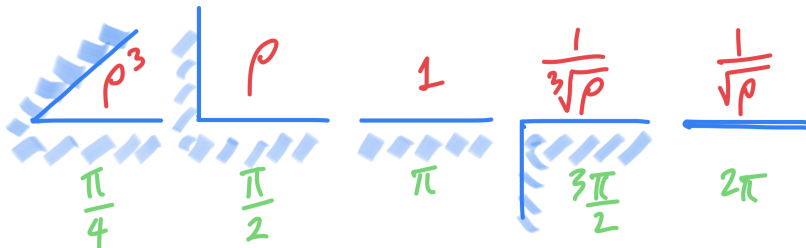
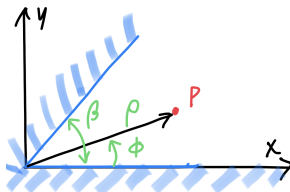


Field at corners and edges in 2D

$$E_\rho \propto \frac{1}{\beta} \rho^{\pi/\beta-1} \sin(\pi\phi/\beta)$$

$$E_\phi \propto \frac{1}{\beta} \rho^{\pi/\beta-1} \cos(\pi\phi/\beta)$$

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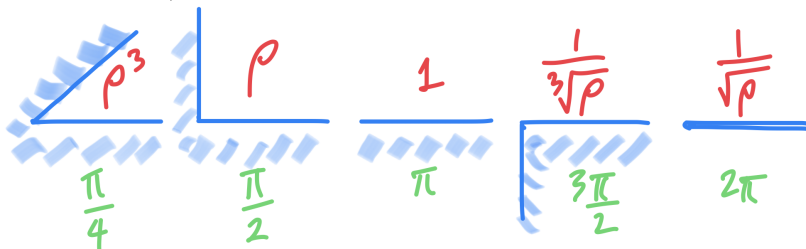
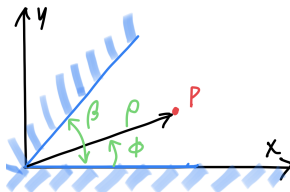
- $E, \sigma \rightarrow 0$ at a deep corner (small β)

Field at corners and edges in 2D

$$E_\rho \propto \frac{1}{\beta} \rho^{\pi/\beta-1} \sin(\pi\phi/\beta)$$

$$E_\phi \propto \frac{1}{\beta} \rho^{\pi/\beta-1} \cos(\pi\phi/\beta)$$

$$\sigma(\rho) \propto \frac{1}{\beta} \rho^{\pi/\beta-1} \cos(\pi\phi/\beta)$$



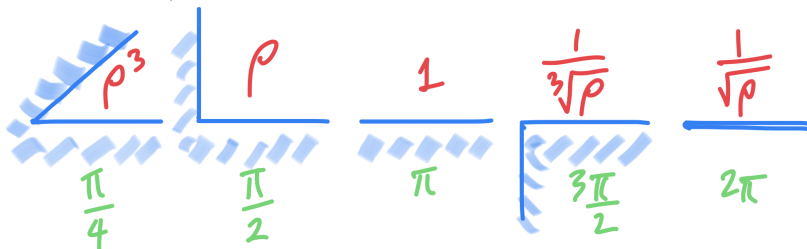
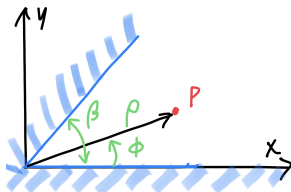
- E, σ independent of ρ for flat surface ($\beta = \pi$)

Field at corners and edges in 2D

$$E_\rho \propto \frac{1}{\beta} \rho^{\pi/\beta-1} \sin(\pi\phi/\beta)$$

$$E_\phi \propto \frac{1}{\beta} \rho^{\pi/\beta-1} \cos(\pi\phi/\beta)$$

$$\sigma(\rho) \propto \frac{1}{\beta} \rho^{\pi/\beta-1} \cos(\pi\phi/\beta)$$



- E, σ blow up for outside corner ($\beta > \pi$)

Finite element analysis (FEA) for electrostatics

- Numerical solution to BVP in 2D region R with Dirichlet bc on boundary curve C

$$\nabla^2 \psi = -g$$

- For a piecewise continuous *test function* $\phi(x, y)$ that vanishes on C

$$\int_R \phi \left(\nabla^2 \psi + g \psi \right) dx dy = 0$$

- Using Green's 1st identity

$$\int_R (\vec{\nabla} \phi \cdot \vec{\nabla} \psi + \phi \nabla^2 \psi) dx dy = \oint_C \phi \frac{\partial \psi}{\partial n} da = 0$$

- we get

$$\int_R (\vec{\nabla} \phi \cdot \vec{\nabla} \psi - g \phi) dx dy = 0$$

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- Test function ϕ with

$$\int_R (\vec{\nabla} \phi \cdot \vec{\nabla} \psi - g \phi) \, dx \, dy = 0$$

- Galerkin's method involves localized, linearly independent functions ϕ_{ij} within (overlapping) cells in R around points (x_i, y_j) to form an expansion for the solution

$$\psi(x, y) \simeq \sum_{ij} \Psi_{ij} \phi_{ij}(x, y)$$

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- Example, piecewise bilinear functions on square lattice with spacing h

$$\phi_{ij}(x, y) = \begin{cases} \left(1 - \frac{|x-x_i|}{h}\right) \left(1 - \frac{|y-y_j|}{h}\right), & |x-x_i|, |y-y_j| \leq h \\ 0, & \text{elsewhere} \end{cases}$$

$$\rightarrow \psi(x_i, y_j) \simeq \Psi_{ij}$$

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- Setting $\phi = \phi_{ij}$ for every internal lattice site (i, j)

$$\begin{aligned} \int_R (\vec{\nabla} \phi_{ij} \cdot \vec{\nabla} \psi - g \phi_{ij}) \, dx \, dy &= 0 \\ \sum_{k,l} \Psi_{kl} \int_R (\vec{\nabla} \phi_{ij} \cdot \vec{\nabla} \phi_{kl}) \, dx \, dy &= \int_R g \phi_{ij} \, dx \, dy \end{aligned}$$

Finite element analysis (FEA) for electrostatics

- Equation with functionals of local basis functions

$$\sum_{k,l} \psi_{kl} \int_R \vec{\nabla} \phi_{ij}(x, y) \cdot \vec{\nabla} \phi_{kl}(x, y) dx dy = \int_R g(x, y) \phi_{ij}(x, y) dx dy$$

- If $g(x, y)$ varies slowly around each site (fine enough lattice spacing to match the source geometry)

$$\sum_{k,l} \psi_{kl} \int_R \vec{\nabla} \phi_{ij}(x, y) \cdot \vec{\nabla} \phi_{kl}(x, y) dx dy = g(x_i, y_j) \int_R \phi_{ij}(x, y) dx dy$$

- The integrals depend only on the basis functions and can be pre-evaluated for a given mesh
- leading to a set of linear algebraic equations for the expansion coefficients ψ_{kl}
- involving a large but sparse coefficient matrix since the basis functions ϕ_{ij} vanish outside a small cell

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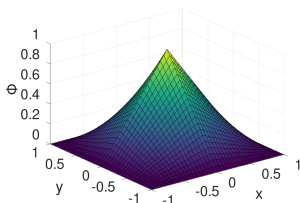
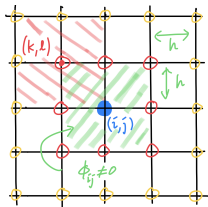
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Example: square lattice with "pyramid" basis



$$\phi_{i,j}(x,y) = \left(1 - \frac{|x - x_i|}{h}\right) \left(1 - \frac{|y - y_j|}{h}\right), \quad |x - x_i|, |y - y_j| \leq h$$

$$\int_R \phi_{i,j} \, dx \, dy = h^2, \quad \int_R \vec{\nabla} \phi_{i,j} \cdot \vec{\nabla} \phi_{i,j} \, dx \, dy = \frac{8}{3}$$

$$\int_R \vec{\nabla} \phi_{i,j} \cdot \vec{\nabla} \phi_{i\pm 1,j} \, dx \, dy = \int_R \vec{\nabla} \phi_{i,j} \cdot \vec{\nabla} \phi_{i,j\pm 1} \, dx \, dy = -\frac{1}{3}$$

$$\int_R \vec{\nabla} \phi_{i,j} \cdot \vec{\nabla} \phi_{i\pm 1,j\pm 1} \, dx \, dy = -\frac{1}{3}$$

$$\int_R \vec{\nabla} \phi_{i,j} \cdot \vec{\nabla} \phi_{k,l} \, dx \, dy = 0, \quad |k - i| > 1 \text{ or } |l - j| > 1$$

Finite element analysis

- Square lattice isn't optimal in many cases
- We need the mesh spacing to be locally adapted to the rate of change of the solution ($|\vec{\nabla}\psi|$)
- using a simple generic shape that can be resized easily
- Triangular elements most popular (and tetrahedral cells in 3D)

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Laplace eqn in spherical coordinates

$$\frac{1}{r} \partial_r^2 (r\Phi) + \frac{1}{r^2 \sin \theta} (\sin \theta \partial_\theta \Phi) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 \Phi = 0$$

- Separation of variables

$$\Phi = \frac{U(r)}{r} P(\theta) Q(\phi), \quad \frac{(r \sin \theta)^2}{UPQ} \nabla^2 \Phi = 0$$

$$(r \sin \theta)^2 \left[\frac{1}{U} \frac{d^2 U}{dr^2} + \frac{1}{r^2 P} \frac{d}{d\theta} \left(\sin \theta \frac{dP}{d\theta} \right) \right] + \frac{1}{Q} \frac{d^2 Q}{d\phi^2} = 0$$

$$\frac{1}{Q} \frac{d^2 Q}{d\phi^2} = -m^2 \rightarrow Q(\phi) = e^{\pm im\phi}$$

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dP}{d\theta} \right) + \left[l(l+1) - \left(\frac{m}{\sin \theta} \right)^2 \right] P = 0$$

$$\frac{d^2 U}{dr^2} - \frac{l(l+1)}{r^2} U = 0 \rightarrow U(r) = A r^{l+1} + B r^{-l}$$

Legendre polynomials

- Equation for P recast in terms of $x = \cos \theta$ gives *generalized Legendre eqn*

$$\frac{d}{dx} \left[(1 - x^2) \frac{dP}{dx} \right] + \left[l(l + 1) - \frac{m^2}{1 - x^2} \right] P = 0$$

- with the solutions called *associated Legendre functions*
- simplifies to Legendre eqn for $m = 0$

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Legendre polynomials $P_l(x)$

$$P_l(x) = \frac{1}{2^l l!} \left(\frac{d}{dx} \right)^l (x^2 - 1)^l$$

$$P_0(x) = 1$$

$$P_1(x) = x$$

$$P_2(x) = (3x^2 - 1)/2$$

$$P_3(x) = (5x^3 - 3x)/2$$

$$P_4(x) = (35x^4 - 30x^2 + 3)/8$$

$$P_5(x) = (63x^5 - 70x^3 + 15x)/8$$

$$P_6(x) = (231x^6 - 315x^4 + 105x^2 - 5)/16$$

$$P_7(x) = (429x^7 - 693x^5 + 315x^3 - 35x)/16$$

Legendre polynomials

$$P_l(1) = 1$$
$$\int_{-1}^1 P_l(x) P_{l'}(x) dx = \delta_{ll'} \left(\frac{2}{2l+1} \right)$$

- $P_n(x)$ has n simple roots in $[-1, 1]$ symmetric around 0
- The Legendre polynomials form a complete set of functions over the interval $[-1, 1]$ with weighting function 1, that is any well-behaved function $f(x)$ can be written as

$$f(x) = \sum_k c_k P_k(x)$$

with a uniquely determined set of coefficients $\{c_k\}$.

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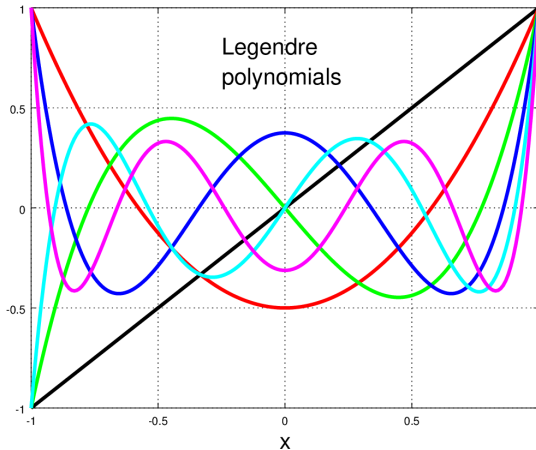
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Azimuthal symmetry ($m = 0$)

- General solution

$$\Phi(r, \theta) = \sum_{l=0}^{\infty} \left[A_l r^l + B_l r^{-(l+1)} \right] P_l(\cos \theta)$$

- with $\{A_l, B_l\}$ to be determined from bc
- Example, $\Phi(a, \theta) = V(\theta)$ specified on the surface of a sphere
 - $B_l = 0$ for finite Φ at $r = 0$
 - bc is just Legendre expansion of $V(\theta)$

$$V(\theta) = \sum_{l=0}^{\infty} A_l a^l P_l(\cos \theta)$$

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Example

- Sphere with hemispheres at $\pm V$

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- Potential inside the sphere

$$\begin{aligned} \Phi(r, \theta) &= V \frac{r}{a} \left[\frac{3}{2} P_1(\cos \theta) - \frac{7}{8} \left(\frac{r}{a} \right)^2 P_3(\cos \theta) \right. \\ &\quad \left. + \frac{11}{16} \left(\frac{r}{a} \right)^4 P_5(\cos \theta) + \dots \right] \\ &= V \frac{r}{a} \cos \theta \left[\frac{3}{2} - \frac{7}{16} \left(\frac{r}{a} \right)^2 (5 \cos^2 \theta - 3) + \dots \right] \end{aligned}$$

Example

- Sphere with hemispheres at $\pm V$

$$V(\theta) = \begin{cases} V, & 0 \leq \theta < \pi/2 \\ -V, & \pi/2 < \theta \leq \pi \end{cases}$$

- For potential outside the sphere,

$$\left(\frac{r}{a}\right)^l \rightarrow \left(\frac{a}{r}\right)^{l+1}$$
$$\Phi(r, \theta) = V \left(\frac{a}{r}\right)^2 \cos \theta \left[\frac{3}{2} - \frac{7}{16} \left(\frac{a}{r}\right)^2 (5 \cos^2 \theta - 3) + \dots \right]$$

Azimuthal symmetry ($m = 0$)

- Consider the general solution

$$\Phi(r, \theta) = \sum_{l=0}^{\infty} \left[A_l r^l + B_l r^{-(l+1)} \right] P_l(\cos \theta)$$

- which reduces, on the axis of symmetry $r = z$, to

$$\Phi(r, \theta = 0) = \sum_{l=0}^{\infty} \left[A_l r^l + B_l r^{-(l+1)} \right]$$

- The series expansion of the solution $\Phi(z = r)$ is unique
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- If we know the expansion along the z -axis, the general solution can be constructed by simply attaching a factor of $P_l(\cos \theta)$ to every power of r^l and $r^{-(l+1)}$

Conducting sphere with $\pm V$ on hemispheres – again

- Starting with the on-axis solution

$$\Phi(z=r) = V \left[1 - \frac{r^2 - a^2}{r\sqrt{r^2 + a^2}} < 1 \right]$$

- we construct the series expansion

$$\frac{\Phi(z=r)}{V} = 1 - \frac{1 - \alpha}{\sqrt{1 + \alpha^2}}, \quad \alpha = \left(\frac{a}{r}\right)^2$$

- and place a P_{2k-1} next to each $\alpha^k \propto r^{-2k}$

$$\frac{\Phi(z=r)}{V} = \frac{3}{2}a^2r^{-2} - \frac{7}{8}a^4r^{-4} + \frac{11}{16}a^6r^{-6} + \dots$$

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$$\Phi(r, \theta) = V \left(\frac{a}{r}\right)^2 \left[\frac{3}{2}P_1(\cos \theta) - \frac{7}{8}\left(\frac{a}{r}\right)^2 P_3(\cos \theta) + \dots \right]$$

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$$\Phi(r, \theta) = V \left(\frac{a}{r}\right)^2 \cos \theta \left[\frac{3}{2} - \frac{7}{16}\left(\frac{a}{r}\right)^2 (5\cos^2 \theta - 3) + \dots \right]$$

Legendre expansion of $1/|\vec{x} - \vec{x}'|$

- The potential at point $\vec{x}$ due to a point charge at $\vec{x}'$ appears in the Green function

$$\frac{1}{|\vec{x} - \vec{x}'|} = \left(|\vec{x}|^2 + |\vec{x}'|^2 - 2|\vec{x}||\vec{x}'|\cos\gamma \right)^{-1/2}$$

- If we choose the z-axis along $\vec{x}'$, $\theta = \gamma$, azimuthal symmetry means

$$\frac{1}{|\vec{x} - \vec{x}'|} = \sum_{l=0}^{\infty} \left[A_l r^l + B_l r^{-(l+1)} \right] P_l(\cos\gamma), \quad \vec{x} \neq \vec{x}'$$

- For $\vec{x}$ on the z-axis,

$$\frac{1}{|\vec{x} - \vec{x}'|} = \frac{1}{|r - r'|} = \frac{1}{r_{>}} \sum_{l=0}^{\infty} \left(\frac{r_{<}}{r_{>}} \right)^l$$

where $r_{<}$, $r_{>}$ are the smaller, larger of $|\vec{x}|$ and $|\vec{x}'|$

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- We put in the P_l 's for the general case

$$\frac{1}{|\vec{x} - \vec{x}'|} = \sum_{l=0}^{\infty} \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos \gamma)$$

General solution in spherical coordinates ($m \neq 0$)

$$\frac{d}{dx} \left[(1-x^2) \frac{dP}{dx} \right] + \left[l(l+1) - \frac{m^2}{1-x^2} \right] P_l^m(x) = 0$$

- Solutions called associated Legendre functions with integer $l \geq 0$, $|m| \leq l$
- For $m > 0$

$$P_l^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m}{dx^m} P_l(x)$$

- For negative m

$$P_l^{-m}(x) = (-1)^m \frac{(l-m)!}{(l+m)!} P_l^m(x)$$

- Also, for all m

$$P_l^m(x) = \frac{(-1)^m}{2^l l!} (1-x^2)^{m/2} \frac{d^{l+m}}{dx^{l+m}} (x^2-1)^l$$

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Associated Legendre functions

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$$P_l^m(x) = \frac{(-1)^m}{2^l l!} (1 - x^2)^{m/2} \frac{d^{l+m}}{dx^{l+m}} (x^2 - 1)^l, \quad l \geq 0, \quad |m| \leq l$$

For fixed m , P_l^m form an orthogonal set in $x \in [-1, 1]$

$$\int_{-1}^1 P_l^m(x) P_{l'}^m(x) dx = \frac{2}{2l+1} \frac{(l-m)!}{(l+m)!} \delta_{ll'}$$

Spherical harmonics

- $P_l^m(\theta)$, $Q_m(\phi)$ from separation of variables for Φ
- For fixed m , P_l^m form an orthogonal set in $x \in [-1, 1]$

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- $Q_m(\phi) = e^{im\phi}$ form a complete orthogonal set in $\phi \in [0, 2\pi]$

$$\int_0^{2\pi} Q_m^*(\phi) Q_{m'}(\phi) d\phi = \int_0^{2\pi} e^{i(m'-m)\phi} d\phi = 2\pi \delta_{mm'}$$

- The product forms a complete orthogonal set on the surface of the unit sphere

$$Y_{lm}(\theta, \phi) = \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta) e^{im\phi}$$

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Spherical harmonics

- Definition

$$Y_{lm}(\theta, \phi) = \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta) e^{im\phi}$$

$$Y_{l,-m}(\theta, \phi) = (-1)^m Y_{lm}^*(\theta, \phi)$$

$$Y_{l0}(\theta, \phi) = \sqrt{\frac{2l+1}{4\pi}} P_l(\cos \theta)$$

- Orthonormality

$$\int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta Y_{l'm'}^*(\theta, \phi) Y_{lm}(\theta, \phi) = \delta_{l'l} \delta_{m'm}$$

- Completeness

$$\sum_{l=0}^{\infty} \sum_{m=-l}^l Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) = \delta(\phi - \phi') \delta(\cos \theta - \cos \theta')$$

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Spherical harmonics

$$Y_{lm} = \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta) e^{im\phi}$$

$$Y_{00} = \frac{1}{\sqrt{4\pi}}$$

$$Y_{10} = \sqrt{\frac{3}{4\pi}} \cos \theta$$

$$Y_{11} = -\sqrt{\frac{3}{8\pi}} \sin \theta e^{i\phi}$$

$$Y_{20} = \sqrt{\frac{5}{4\pi}} \left(\frac{3}{2} \cos^2 \theta - \frac{1}{2} \right)$$

$$Y_{21} = -\sqrt{\frac{15}{8\pi}} \sin \theta \cos \theta e^{i\phi}$$

$$Y_{22} = \frac{1}{4} \sqrt{\frac{15}{2\pi}} \sin^2 \theta e^{i2\phi}$$

$$Y_{30} = \sqrt{\frac{7}{4\pi}} \left(\frac{5}{2} \cos^3 \theta - \frac{3}{2} \cos \theta \right)$$

$$Y_{31} = -\frac{1}{4} \sqrt{\frac{21}{4\pi}} \sin \theta (5 \cos^2 \theta - 1) e^{i\phi}$$

$$Y_{32} = \frac{1}{4} \sqrt{\frac{105}{2\pi}} \sin^2 \theta \cos \theta e^{i2\phi}$$

$$Y_{33} = -\frac{1}{4} \sqrt{\frac{35}{4\pi}} \sin^3 \theta e^{i3\phi}$$

Spherical harmonics

- Arbitrary (well-behaved) function $g(\theta, \phi)$ can be expanded as

$$g(\theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} Y_{lm}(\theta, \phi)$$

$$A_{lm} = \int d\Omega Y_{lm}^*(\theta, \phi) g(\theta, \phi)$$

$$g(\theta = 0, \phi) = \sum_{l=0}^{\infty} \sqrt{\frac{2l+1}{4\pi}} A_{l0}$$

$$A_{l0} = \sqrt{\frac{2l+1}{4\pi}} \int d\Omega P_l(\cos \theta) g(\theta, \phi)$$

- General solution for the potential

$$\Phi(r, \theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l \left[A_{lm} r^l + B_{lm} r^{-(l+1)} \right] Y_{lm}(\theta, \phi)$$

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- Legendre polynomials can be expressed as a superposition of spherical harmonics

$$P_l(\cos \gamma) = \frac{4\pi}{2l+1} \sum_{m=-l}^l Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi)$$
$$\cos \gamma = \cos \theta' \cos \theta + \sin \theta' \sin \theta \cos(\phi - \phi')$$

- which leads to

$$\frac{1}{|\vec{x} - \vec{x}'|} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi)$$

Addition theorem

- Legendre polynomials can be expressed as a superposition of spherical harmonics

$$P_l(\cos \gamma) = \frac{4\pi}{2l+1} \sum_{m=-l}^l Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi)$$
$$\cos \gamma = \cos \theta' \cos \theta + \sin \theta' \sin \theta \cos(\phi - \phi')$$

- which leads to

$$\frac{1}{|\vec{X} - \vec{X}'|} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi)$$

Laplace eqn in cylindrical coordinates

$$\partial_\rho^2 \Phi + \frac{1}{\rho} \partial_\rho \Phi + \frac{1}{\rho^2} \partial_\phi^2 \Phi + \partial_z^2 \Phi = 0$$

- Separation of variables

$$\Phi = R(r) Q(\phi) Z(z), \quad \frac{1}{\Phi} \nabla^2 \Phi = 0$$

$$\frac{1}{R} \left(\frac{d^2 R}{d\rho^2} + \frac{1}{\rho} \frac{dR}{d\rho} \right) + \frac{1}{\rho^2} \frac{1}{Q} \frac{d^2 Q}{d\phi^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = 0$$

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$$\frac{d^2 R}{dx^2} + \frac{1}{x} \frac{dR}{dx} + \left(1 - \frac{\nu^2}{x^2} \right) R = 0$$

Solutions called Bessel functions (of order ν)

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Solutions called Bessel functions (of order ν)

- Power series for Bessel functions of the 1st kind

$$R(x) = x^\alpha \sum_{j=0}^{\infty} a_j x^j \rightarrow a_{2j} = \frac{(-1)^j \Gamma(\alpha + 1)}{2^{2j} j! \Gamma(j + \alpha + 1)} a_0$$

$$J_\nu(x) = \left(\frac{x}{2}\right)^\nu \sum_{j=0}^{\infty} \frac{(-1)^j}{j! \Gamma(j + \nu + 1)} \left(\frac{x}{2}\right)^{2j}$$

$$J_{-\nu}(x) = \left(\frac{x}{2}\right)^{-\nu} \sum_{j=0}^{\infty} \frac{(-1)^j}{j! \Gamma(j - \nu + 1)} \left(\frac{x}{2}\right)^{2j}$$

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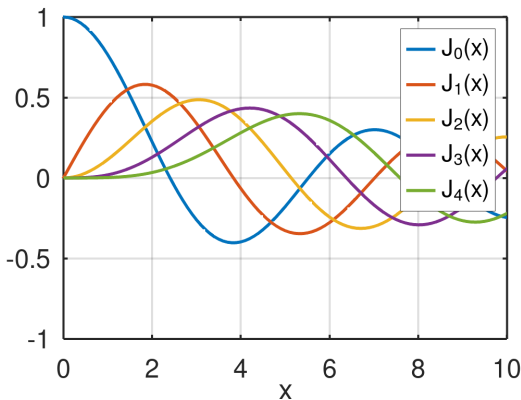
$$N_\nu(x) = \frac{J_\nu(x) \cos \nu\pi - J_{-\nu}(x)}{\sin \nu\pi}$$

- Hankel functions (Bessel functions of the 3rd kind) defined as another independent set

$$H_\nu^{(1)}(x) = J_\nu(x) + iN_\nu(x)$$

$$H_\nu^{(2)}(x) = J_\nu(x) - iN_\nu(x)$$

Bessel functions



$$J_m(x \rightarrow 0) \approx \frac{1}{m!} \left(\frac{x}{2}\right)^m$$

Roots

$$J_m(x_{mn}) = 0$$

$$x_{0n} \simeq 2.504, 5.520, 8.654, \dots$$

$$x_{1n} \simeq 3.832, 7.016, 10.173, \dots$$

$$x_{2n} \simeq 5.136, 8.417, 11.620, \dots$$

$$x_{3n} \simeq 5.136, 8.417, 11.620, \dots$$

$$x_{4n} \simeq 7.588, 11.065, 14.373, \dots$$

- $J_\nu(x)$ are finite at both $x = 0$ and $x \rightarrow \infty$

$$J_\nu(x \ll 1) \rightarrow \frac{1}{\Gamma(\nu + 1)} \left(\frac{x}{2}\right)^\nu$$

$$J_\nu(x \gg 1, \nu) \rightarrow \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{\nu\pi}{2} - \frac{\pi}{4}\right)$$

- $N_\nu(x)$ blow up at $x = 0$

$$N_\nu(x \ll 1) \rightarrow \begin{cases} \frac{2}{\pi} \left[\ln\left(\frac{x}{2}\right) + 0.5772 + \dots \right], & \nu = 0 \\ -\frac{\Gamma(\nu)}{\pi} \left(\frac{2}{x}\right)^\nu, & \nu \neq 0 \end{cases}$$

$$N_\nu(x \gg 1, \nu) \rightarrow \sqrt{\frac{2}{\pi x}} \sin\left(x - \frac{\nu\pi}{2} - \frac{\pi}{4}\right)$$

- Bessel functions all satisfy the recursion relations

$$\Omega_{\nu-1} + \Omega_{\nu+1} = \frac{2\nu}{x} \Omega_{\nu}(x)$$

$$\Omega_{\nu-1} - \Omega_{\nu+1} = 2 \frac{d\Omega_{\nu}(x)}{dx}$$

where Ω_{ν} is one of J_{ν} , N_{ν} and H_{ν}

Radial solution in cylindrical coordinates

- $\sqrt{\rho} J_\nu \left(x_{\nu n} \frac{\rho}{a} \right)$ for fixed ν form an orthogonal set in the interval $\rho \in [0, a]$

$$\int_0^a d\rho \rho J_\nu \left(x_{\nu n} \frac{\rho}{a} \right) J_\nu \left(x_{\nu n'} \frac{\rho}{a} \right) = \frac{a^2}{2} J_{\nu+1}^2(x_{\nu n}) \delta_{nn'}$$

- and any $f(\rho)$ can be expanded in $[0, a]$ as

$$f(\rho) = \sum_{n=1}^{\infty} A_{\nu n} J_\nu \left(x_{\nu n} \frac{\rho}{a} \right)$$
$$A_{\nu n} = \frac{2}{a^2 J_{\nu+1}^2(x_{\nu n})} \int_0^a d\rho \rho f(\rho) J_\nu \left(x_{\nu n} \frac{\rho}{a} \right)$$

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$$\partial_\rho^2 \Phi + \frac{1}{\rho} \partial_\rho \Phi + \frac{1}{\rho^2} \partial_\phi^2 \Phi + \partial_z^2 \Phi = 0$$

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- Solutions are **modified** Bessel functions

$$I_\nu(x) = i^{-\nu} J_\nu(ix)$$

$$K_\nu(x) = \frac{\pi}{2} i^{\nu+1} H_\nu^{(1)}(ix)$$

Modified Bessel functions

- $I_\nu(x)$ are finite at $x = 0$ and blow up at $x \rightarrow \infty$

$$I_\nu(x \ll 1) \rightarrow \frac{1}{\Gamma(\nu + 1)} \left(\frac{x}{2}\right)^\nu$$

$$I_\nu(x \gg 1, \nu) \rightarrow \sqrt{\frac{2}{\pi x}} e^x \left[1 + \mathcal{O}\left(\frac{1}{x}\right) \right]$$

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- Cylinder with radius a with bottom/top at $z = 0/L$
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$$\Phi(\rho, \phi, z) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} J_m(k_{mn}\rho) \sinh(k_{mn}z) (A_{mn} \cos m\phi$$

$$+ B_{mn} \sin m\phi)$$

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- with coefficients (extra 1/2 for $m = 0$)

$$A_{mn} = \frac{2 \operatorname{cosech}(k_{mn}L)}{\pi a^2 J_{m+1}^2(k_{mn}a)} \int_0^{2\pi} d\phi \int_0^a d\rho \rho V(\rho, \phi) J_m(k_{mn}\rho) \sin m\phi$$

$$B_{mn} = \frac{2 \operatorname{cosech}(k_{mn}L)}{\pi a^2 J_{m+1}^2(k_{mn}a)} \int_0^{2\pi} d\phi \int_0^a d\rho \rho V(\rho, \phi) J_m(k_{mn}\rho) \cos m\phi$$

3D BVP in cylindrical coordinates

- Compact boundary at $\rho = a$ gives discrete k values and a corresponding Fourier-Bessel series for $\Phi/Q(\phi)$ with terms like $J(k\rho) \sinh(kz)$ or $I(k\rho) \sin(kz)$

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- gives simple Fourier series in ϕ with

$$\left. \begin{matrix} A_m(k) \\ B_m(k) \end{matrix} \right\} = \frac{k}{\pi} \int_0^\infty d\rho \rho \int_0^{2\pi} d\phi V(\rho, \phi) J_m(k\rho) \left\{ \begin{matrix} \sin m\phi \\ \cos m\phi \end{matrix} \right.$$

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$$\begin{aligned} G(\vec{x}, \vec{x}') &= \frac{1}{|\vec{x} - \vec{x}'|} - \frac{a}{r'} \frac{1}{\left| \vec{x} - \left(\frac{a}{r'}\right)^2 \vec{x}' \right|} \\ &= 4\pi \sum_{l,m} \left[\frac{r_{<}^l}{r_{>}^{l+1}} - \frac{a}{r'} \frac{(a^2/r')^l}{r^{l+1}} \right] Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) \end{aligned}$$

Green Functions in Spherical Coordinates

- With $\rho(\vec{x}') \neq 0$, we need $G(\vec{x}, \vec{x}')$ for the solution, preferably in terms of basis functions in the appropriate coordinates
- Free-space version in spherical harmonics

$$\frac{1}{|\vec{x} - \vec{x}'|} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi)$$

- For the exterior problem with boundary at $r = a$ ($r, r' > a$)

$$\begin{aligned} G(\vec{x}, \vec{x}') &= \frac{1}{|\vec{x} - \vec{x}'|} - \frac{a}{r'} \frac{1}{\left| \vec{x} - \left(\frac{a}{r'}\right)^2 \vec{x}' \right|} \\ &= 4\pi \sum_{l,m} \left[\frac{r_{<}^l}{r_{>}^{l+1}} - \frac{a}{r'} \frac{(a^2/r')^l}{r^{l+1}} \right] Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) \\ &= 4\pi \sum_{l,m} \left[\frac{r_{<}^l}{r_{>}^{l+1}} - \frac{1}{a} \left(\frac{a^2}{rr'} \right)^{l+1} \right] Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) \end{aligned}$$

Green Functions in Spherical Coordinates

- Exterior problem with boundary at $r = a$ ($r, r' > a$)

$$G(\vec{x}, \vec{x}') = 4\pi \sum_{l,m} \left[\frac{r_{<}^l}{r_{>}^{l+1}} - \frac{1}{a} \left(\frac{a^2}{rr'} \right)^{l+1} \right] Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi)$$

- Radial part

$$\left[\frac{r_{<}^l}{r_{>}^{l+1}} - \frac{1}{a} \left(\frac{a^2}{rr'} \right)^{l+1} \right] = \begin{cases} \frac{1}{r'^{l+1}} \left(r^l - \frac{a^{2l+1}}{r^{l+1}} \right), & r < r' \\ \left(r'^l - \frac{a^{2l+1}}{r'^{l+1}} \right) \frac{1}{r^{l+1}}, & r > r' \end{cases}$$

- shows

$$G(r = a) = G(r' = a) = 0, \quad G(r \rightarrow \infty) \rightarrow 0$$

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Green Functions in Spherical Coordinates

- Dirichlet Green fn for region bounded by concentric spheres at $r = a, b$

$$G(\vec{x}, \vec{x}') = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi)}{(2l+1) \left[1 - \left(\frac{a}{b}\right)^{2l+1} \right]} \\ \times \left(r_{<}^l - \frac{a^{2l+1}}{r_{<}^{l+1}} \right) \left(\frac{1}{r_{>}^{l+1}} - \frac{r_{>}^l}{b^{l+1}} \right)$$

- satisfies

$$\nabla^2 G(\vec{x}, \vec{x}') = -4\pi \delta(\vec{x} - \vec{x}')$$

- and can be used to calculate the general solution

$$\Phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int_V \rho(\vec{x}') G(\vec{x}, \vec{x}') d^3x' - \frac{1}{4\pi} \int_S \Phi(\vec{x}') \frac{\partial G(\vec{x}, \vec{x}')}{\partial n'} da'$$

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Eigenfunction expansions

- The differential equation

$$\nabla^2 \psi(\vec{x}) + [f(\vec{x}) + \lambda] \psi(\vec{x}) = 0$$

- has well-behaved (finite, continuous) solutions $\psi_n(\vec{x})$ (called the *eigenfunctions*) in some region V for only certain values λ_n of λ (called the *eigenvalues*)

$$\nabla^2 \psi_n(\vec{x}) + [f(\vec{x}) + \lambda_n] \psi_n(\vec{x}) = 0$$

- The eigenfunctions are orthogonal and can be normalized so that

$$\int_V \psi_n^*(\vec{x}) \psi_m(\vec{x}) d^3x = \delta_{nm}$$

- and we'll assume that they form a complete set

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$$-\frac{\hbar^2}{2m} \nabla^2 \psi_n(\vec{x}) + V(\vec{x}) \psi_n(\vec{x}) = E_n \psi_n(\vec{x})$$

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- which has the same form with

$$f(\vec{x}) = -\frac{2m}{\hbar^2} V(\vec{x}), \quad \lambda_n = \frac{2m}{\hbar^2} E_n$$

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- The corresponding Green function for $\lambda \notin \{\lambda_n\}$,

$$\nabla_x^2 G(\vec{x}, \vec{x}') + [f(\vec{x}) + \lambda] G(\vec{x}, \vec{x}') = -4\pi \delta(\vec{x} - \vec{x}')$$

- with the same boundary conditions, can be expanded as

$$G(\vec{x}, \vec{x}') = \sum_k a_k(\vec{x}') \psi_k(\vec{x}), \quad (\nabla^2 + f) \psi_k = -\lambda_k \psi_k$$

- Plugging the expansion into the equation,

$$\left[\nabla_x^2 + f(\vec{x}) + \lambda \right] \sum_k a_n(\vec{x}') \psi_k(\vec{x}) = -4\pi \delta(\vec{x} - \vec{x}')$$

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Eigenfunction expansions – example

- The wave equation in free space (no boundaries)

$$\left(\nabla^2 + k^2\right) \psi_{\vec{k}}(\vec{x}) = 0$$

- has a continuum of eigenvalues k^2 with plane-wave eigenfunctions

$$\psi_{\vec{k}}(\vec{x}) = \frac{1}{(2\pi)^{3/2}} e^{i\vec{k}\cdot\vec{x}}$$

- and delta-function normalization

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Eigenfunction expansions – example

- Same problem with Dirichlet boundary conditions on rectangular box with diagonal from $\vec{0}$ to (a, b, c)
- has eigenfunctions, eigenvalues given by

$$(\nabla^2 + k_{lmn}^2) \psi_{lmn}(x, y, z) = 0$$

$$\psi_{lmn}(x, y, z) = \sqrt{\frac{8}{abc}} \sin\left(\frac{l\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) \sin\left(\frac{n\pi z}{c}\right)$$

$$k_{lmn}^2 = \pi^2 \left[\left(\frac{l}{a}\right)^2 + \left(\frac{m}{b}\right)^2 + \left(\frac{n}{c}\right)^2 \right]$$

- and the Green function expansion is

$$G(\vec{x}, \vec{x}') = \frac{32\pi}{abc} \sum_{l,m,n=0}^{\infty} \frac{1}{k_{lmn}^2} \sin\left(\frac{l\pi x}{a}\right) \sin\left(\frac{l\pi x'}{a}\right) \\ \times \sin\left(\frac{m\pi y}{b}\right) \sin\left(\frac{m\pi y'}{b}\right) \sin\left(\frac{n\pi z}{c}\right) \sin\left(\frac{n\pi z'}{c}\right)$$

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$$\psi_{lmn}(x, y, z) = \sqrt{\frac{8}{abc}} \sin\left(\frac{l\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) \sin\left(\frac{n\pi z}{c}\right)$$

$$k_{lmn}^2 = \pi^2 \left[\left(\frac{l}{a}\right)^2 + \left(\frac{m}{b}\right)^2 + \left(\frac{n}{c}\right)^2 \right]$$

- and the Green function expansion is

$$G(\vec{x}, \vec{x}') = \frac{32\pi}{abc} \sum_{l,m,n=0}^{\infty} \frac{1}{k_{lmn}^2} \sin\left(\frac{l\pi x}{a}\right) \sin\left(\frac{l\pi x'}{a}\right) \\ \times \sin\left(\frac{m\pi y}{b}\right) \sin\left(\frac{m\pi y'}{b}\right) \sin\left(\frac{n\pi z}{c}\right) \sin\left(\frac{n\pi z'}{c}\right)$$

Eigenfunction expansions – example

- Same problem with Dirichlet boundary conditions on rectangular box with diagonal from $\vec{0}$ to (a, b, c)
- has eigenfunctions, eigenvalues given by

$$(\nabla^2 + k_{lmn}^2) \psi_{lmn}(x, y, z) = 0$$

$$\psi_{lmn}(x, y, z) = \sqrt{\frac{8}{abc}} \sin\left(\frac{l\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) \sin\left(\frac{n\pi z}{c}\right)$$

$$k_{lmn}^2 = \pi^2 \left[\left(\frac{l}{a}\right)^2 + \left(\frac{m}{b}\right)^2 + \left(\frac{n}{c}\right)^2 \right]$$

- and the Green function expansion is

$$G(\vec{x}, \vec{x}') = \frac{32\pi}{abc} \sum_{l,m,n=0}^{\infty} \frac{1}{k_{lmn}^2} \sin\left(\frac{l\pi x}{a}\right) \sin\left(\frac{l\pi x'}{a}\right) \\ \times \sin\left(\frac{m\pi y}{b}\right) \sin\left(\frac{m\pi y'}{b}\right) \sin\left(\frac{n\pi z}{c}\right) \sin\left(\frac{n\pi z'}{c}\right)$$